FCPC: A Dissimilarity-Aware Plug-and-Play Federated Learning Framework for Non-IID IoT Systems

***Abstract*—The proliferation of dual-skewed non-independent and identically distributed (non-IID) data across Internet of Things (IoT) edge devices—characterized by heterogeneous label distributions and imbalanced sample sizes—severely degrades the robustness of conventional federated learning (FL) frameworks. To address this challenge, we propose FCPC, a dissimilarity-aware plug-and-play FL framework. First, we design a novel metric, Jensen-Shannon Divergence and Skew Indicator Number (JSDN), to quantify the degree of data heterogeneity in edge FL systems. Leveraging JSDN, we develop a client pairing algorithm to select client pairs with maximized distribution dissimilarities. Concurrently, we embed a lightweight regularization term into local training to align gradient update directions across paired clients and mitigate dimensional collapse. Extensive experiments on benchmark datasets (CIFAR-10/100, TinyImageNet, MNIST, FEMNIST) and real-world IoT scenarios validate two key advantages of FCPC: 1) It outperforms state-of-the-art baselines by 2%–9.4% under strong heterogeneity (63.17% top-1 accuracy on CIFAR-10 with $\alpha = 0.05$, 9.4 percentage points higher than FedAvg) with negligible computational overhead; 2) As a plug-and-play module, it seamlessly integrates into existing FL algorithms (e.g., FedAvg, FedProx) without modifying their core architectures, yielding a 3.5%–9.4% performance improvement. Overall, this work presents a deployable and compatible solution to enhance FL robustness for practical IoT systems.**

***Index Terms*—Federated learning(FL), Non-IID data, IoT systems, JSDN, Dissimilarity-Aware, Plug-and-Play, FCPC framework**

I. Introduction

In the field of distributed machine learning for Internet of Things (IoT) systems, federated learning (FL) has emerged as a pivotal paradigm. It enables collaborative model training across resource-constrained edge devices while preserving data privacy—a critical requirement for sensitive IoT data (e.g., user behavior logs and industrial operational metrics) [1] and a key enabler for the embodied intelligence that interacts with physical IoT environments [2]. FL has been widely applied in IoT applications, including smart healthcare monitoring, industrial predictive maintenance, and intelligent transportation systems. However, data heterogeneity in IoT environments presents a formidable challenge.

A typical example is the traffic and environmental sensing scenario (Fig. 1), where on-board units, traffic signal controllers, surveillance cameras, and unmanned aerial vehicles (UAVs) generate data characterized by distinct category skew and quantity imbalance (e.g., traffic cameras at urban intersections producing far more samples than those in suburban areas). Such heterogeneity poses notable challenges to latency-sensitive tasks like real-time traffic state prediction and signal timing optimization. Beyond traffic IoT systems, similar data heterogeneity issues persist in other critical industrial tasks reliant on data-driven inference, including load assessment of cylindrical rolling element bearings [3], fault diagnosis in industrial devices via a neuro-fuzzy approach with hypergraph convolution [4], and vibration-based gear wear monitoring and prediction techniques [5]—all of which necessitate robust federated learning frameworks capable of handling non-IID data.

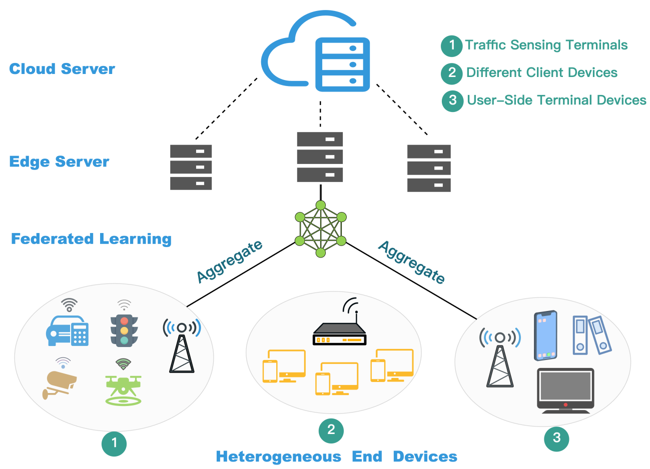

Fig. 1. The non-IID IoT system adopts a cloud-edge-end hierarchical device architecture, which comprises heterogeneous end devices.

Previous research on federated learning has primarily tackled data heterogeneity from three perspectives, each exhibiting limitations in IoT edge scenarios: Firstly, Algorithmic refinements focus on optimizing local training processes [6], [7] or redesigning global aggregation mechanisms [8]. However, many of these approaches impose substantial computational or communication overhead, which is often incompatible with the resource limitations of typical IoT edge hardware. Secondly, Architectural innovations explore advanced model structures (e.g., Vision Transformers [9]) or novel learning paradigms (e.g., SplitFed [10]) to enhance performance. Nevertheless, such methods frequently depend on specific, non-standardized model architectures or require fundamental changes to the federated learning workflow, hindering their seamless adoption in heterogeneous IoT ecosystems. Thirdly, Temporal-spatial explorations investigate patterns like seasonal data fluctuations [11], yet they may not adequately address the static, structural heterogeneity inherent in many IoT deployments, such as persistent label and quantity skews across devices. A critical gap remains: most existing solutions are tailored to mitigate a single type of data skew and lack effective mechanisms to handle the compounded challenges of dual-skewed non-IID data, where both label distribution and sample size vary significantly across clients—a common scenario in real-world IoT systems.

To overcome these limitations, we introduce the Federated Client Pairing and Mutual Constraint (FCPC) framework, motivated by a key observation (detailed in Section 4: Motivation): a potential mutual constraint exists among clients and its strength correlates positively with data distribution dissimilarity. This insight shifts the core challenge to accurately quantifying inter-client data dissimilarity and devising an efficient mechanism to exploit the constraint. FCPC innovates in two aspects. First, it proposes the Jensen-Shannon Divergence and Skew Indicator Number (JSDN) metric, a novel metric that jointly quantifies label and quantity skews. Second, it employs a lightweight JSDN-based client pairing algorithm and injects a pairwise regularization term into local training, actively aligning gradient updates between dissimilar clients. Consequently, FCPC serves as a plug-and-play module that requires no changes to the base FL architecture or aggregation scheme, introduces minimal computational overhead, and maintains orthogonal compatibility—effectively addressing a critical deployment gap in resource-constrained IoT systems.

Contributions. The main contributions of this work are summarized as follows:

- **Novel Insight into Client Constraints:** A mutual constraint relationship is identified among federated learning clients, with its strength correlating positively with data distribution dissimilarity. A lightweight pairwise regularization mechanism is designed to leverage this relationship, mitigating dimensional collapse in non-IID settings—particularly crucial for IoT edge systems.
- **Unified Heterogeneity Quantification Metric:** The JSDN metric is proposed to quantify label and quantity skews within a unified formulation. Providing a more holistic characterization of dual-skewed heterogeneity than single-type metrics, JSDN enables more precise client analysis and pairing.
- **Plug-and-Play Framework for Practical Deployment:** We propose the FCPC framework, which does not require core FL algorithms modification. Its design requires no changes to core FL algorithms, operating as an add-on module that introduces only a client pairing step and a lightweight regularization term, thereby enabling practical deployment in heterogeneous IoT environments with minimal integration cost.

II. Related work

A. Non-IID IoT Systems

The data heterogeneity in federated learning refers to the inconsistent distribution of data across edge devices, violating the independent and identically distributed (IID) assumption [12]. In IoT ecosystems, this phenomenon intensifies due to device-specific deployment environments and sensing capabilities. Current heterogeneity classification encompasses two dimensions:

At the label level, it is manifested as differences in the label space of the data from each participant, which exerts a profound impact on model training [13]. Divergent label spaces across devices degrade model robustness. Industrial IoT deployments exemplify this issue, where vibration sensors in high-risk manufacturing zones report disproportionate “equipment failur” labels, while identical sensors in controlled environments predominantly yield “normal operation” labels.

At the quantity level, this manifests as notable disparities in sample sizes across participants: a single client may possess a vastly larger dataset than the rest, or data volumes may be severely imbalanced among multiple clients.

B. Existing Solutions

To contextualize our approach, we review prevalent strategies for handling non-IID data and analyze their suitability for dual-skewed IoT scenarios.

FedProx [14] is an approach designed to address system heterogeneity and data heterogeneity. It enables flexible local computation, which to some extent mitigates the interference caused by resource-constrained federated participants during the training process. At the same time, it merges a proximal term in the loss function to avoid excessive deviation of client training from the global model.

MOON(Model-Contrastive Learning) [15] mitigates the effects of differences in data distribution by introducing comparative learning techniques into the training process. This method is simple and effective, encouraging model-level feature alignment to improve the performance of federated deep learning models on non-IID datasets.

FastPPO [16] proposes an improved PPO-based scheduling algorithm for personalized federated learning in green IoT. By integrating greedy reward allocation and exponential-decay training duration, it adapts to participants’ dynamic resources and preferences, maximizing FL model accuracy while optimizing energy efficiency and participation willingness.

FedDyn [17] introduces a new approach to construct agent datasets and dynamically extract local knowledge. Instead of using an averaging strategy, it implements focal distillation to emphasize reliable knowledge, effectively solving the problem of knowledge bias.

Summary and Differentiation. As reviewed, the prevailing methods tackle non-IID challenges through diverse strategies: global-model-centric regularization (e.g., FedProx), representation-level contrastive learning (e.g., MOON), dynamic knowledge distillation (e.g., FedDyn), trust evaluation for federated learning in mobile network digital twins (e.g., TFL-DT [18]), local graph-based aggregation for dual skews (e.g., FBLG [19]), causal inference to address aggregation bias (e.g., FedCFA [20]) and reinforcement learning-based personalized scheduling (e.g., FastPPO). While effective in their specific designs, many incur non-negligible computational overhead (e.g., contrastive pairs in MOON, graph construction in FBLG) or require modifications to the core FL aggregation protocol (e.g., VerifyDFL [21], FedDyn, FedCFA), which can hinder their adoption in lightweight, standardized IoT frameworks. Furthermore, except for FBLG which explicitly targets dual skews, most are primarily designed for or evaluated on single-type heterogeneity and lack an intrinsic mechanism to actively leverage inter-client data dissimilarity. In contrast, our proposed FCPC framework is distinguished by its core philosophy of transforming dissimilarity into a convergent force. Specifically, FCPC differs fundamentally by: (1) introducing a lightweight, pairwise constraint that directly utilizes quantified data distribution gaps to align client updates, (2) operating as a plug-and-play module that requires no alteration to the global aggregation scheme and adds minimal computational burden, and (3) employing a unified metric (JSDN) to explicitly target and exploit dual-skewed heterogeneity. This design makes FCPC a uniquely practical and efficient solution tailored for resource-constrained IoT systems where statistical heterogeneity and device limitations coexist.

III. The System Model and Definitions

A. Federated Learning Framework

As a novel machine learning paradigm, federated learning addresses the challenge of training high-performance global models without disclosing local data when multi-source datasets are distributed across different devices or clients. This framework enables model training on decentralized devices (clients) that maintain local datasets without requiring centralized data exchange.

Formally, let $\mathcal{D} = \{D_1, D_2, \ldots, D_n\}$ represent the distributed datasets across n clients, where D_i is the local dataset of client i. The objective of federated learning is to train a global model M by aggregating locally computed model updates while minimizing the loss function $\mathcal{L}$ with respect to model parameters θ. This is expressed mathematically as:

$$\min_{\theta} \mathcal{L}(\theta) = \sum_{i=1}^{n} \frac{|D_i|}{|D|} \mathcal{L}_i(\theta), \tag{1}$$

where $|D_i|$ denotes the number of samples in client i's dataset, $|D| = \sum_{i=1}^{n} |D_i|$ represents the total number of samples across all clients, and $\mathcal{L}_i$ is the loss function computed on the i-th client's dataset. This formulation enables effective feature learning from distributed data while ensuring local data residency and privacy preservation.

FedAvg [22] represents a foundational and widely deployed algorithm in federated learning. In FedAvg:

1) Each client computes model updates based on local data;
2) Client updates are transmitted to the central server;
3) The server aggregates updates via weighted averaging:

$$\theta^{(t+1)} = \theta^{(t)} - \eta \sum_{i=1}^{n} \frac{|D_i|}{|D|} \nabla \mathcal{L}_i(\theta^{(t)}). \tag{2}$$

Where $\theta^{(t)}$ and $\theta^{(t+1)}$ denote the global model parameters updated at time t and after the update, respectively, and η denotes the learning rate that controls the magnitude of the update.

B. Differential Privacy

Differential privacy (DP) [23] provides a rigorous mathematical foundation for privacy preservation in our framework. A randomized mechanism $\mathcal{M}$ satisfies ϵ-differential privacy if: for any two adjacent datasets D and D' differing by at most one element, and any measurable output set $S \subseteq \text{Range}(\mathcal{M})$:

$$\Pr[\mathcal{M}(D) \in S] \leq e^{\epsilon} \cdot \Pr[\mathcal{M}(D') \in S], \tag{3}$$

where $\epsilon > 0$ is the privacy budget governing the privacy-utility tradeoff. The ℓ_1-sensitivity of a function $f : \mathcal{D} \to \mathbb{R}^k$ is defined as:

$$\Delta_1 f = \max_{D \sim D'} \|f(D) - f(D')\|_1, \tag{4}$$

where the maximum is taken over all adjacent datasets. The Laplace mechanism achieves ϵ-DP by adding sensitivity-scaled noise: $\mathcal{M}(D) = f(D) + \text{Lap}(\Delta_1 f/\epsilon)$.

Extending this to federated systems, local differential privacy (LDP) [24] enforces privacy directly at the data source. A mechanism $\mathcal{M}$ satisfies ϵ-LDP if, for all pairs of inputs $v, v' \in \mathcal{D}$ and any output $y \in \mathcal{Y}$:

$$\frac{\Pr[\mathcal{M}(v) = y]}{\Pr[\mathcal{M}(v') = y]} \leq e^{\epsilon}. \tag{5}$$

This client-side protection eliminates server trust assumptions, aligning with federated learning's core principle of data locality.

In our context, consider client k reporting a label distribution vector $h_k \in \mathbb{N}^C$ (where C is the number of classes) and data size $N_k \in \mathbb{N}$. The ℓ_1-sensitivities are $\Delta_1 h_k = 2$ (since changing one sample can affect at most two class counts, e.g., decreasing one count and increasing another) and $\Delta_1 N_k = 1$ (as a single sample change alters the size by ± 1). The LDP perturbation mechanisms are:

$$\tilde{h}_k = h_k + \left(\text{Lap}\left(2/\epsilon_h\right)\right)^C, \tag{6}$$
$$\tilde{N}_k = N_k + \text{Lap}\left(1/\epsilon_N\right), \tag{7}$$

where $\epsilon_h > 0$ and $\epsilon_N > 0$ denote the privacy budgets allocated to the label distribution and size, respectively.

Through sequential composition, the joint mechanism satisfies ϵ-LDP with $\epsilon = \epsilon_h + \epsilon_N$. To minimize the total variance, the optimal privacy budget allocation is:

$$\epsilon_h^* = \epsilon \cdot \frac{\sqrt{C/N_k}}{\sqrt{C/N_k} + 1}, \quad \epsilon_N^* = \epsilon - \epsilon_h^*, \tag{8}$$

where C is the class count and N_k is the local data size. This allocation balances utility between the histogram and scalar protections.

IV. Motivation

A. Dual Skewed Non-IID Data Leads to Worse Global Model Performance Degradation

We conduct a preliminary experiment with the classic FedAvg algorithm as an example. We set up 10 clients, record the classification accuracy in three scenarios in the number of 20 global rounds, and draw a line graph, as shown in Fig. 2. It can be observed that a single label distribution skew or quantity skew has been able to roughly learn the data characteristics in the previous rounds of communication.

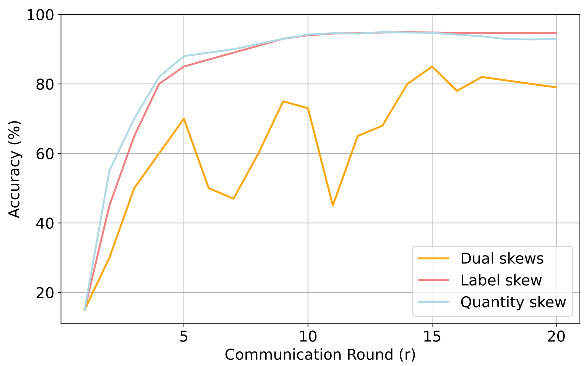

Fig. 2. Impact of dual skewed non-IID data on global model training induced by heterogeneous label distributions among clients and sample size.

Implication 1. *Compared with the single skew, the dual skewed non-IID data has a greater impact on the performance of the global model.*

B. Deep Understanding of Performance Degradation

In federated learning, the differences in data distributions across different clients can lead to dimensional collapse in both global and local models, making it difficult for the model to fully utilize the representation space to distinguish between different classes of data, which in turn affects the performance.

Implication 2. *A bottom-up perspective of dimensional collapse may help minimize the differences in model updates between participants.*

C. Discover Potential Constraints Among Participants

Based on the above study, we consider whether heterogeneous clients can restrict each other to mitigate the low ranking of local models during training. To this end, we conducted simple preliminary experiments. Specifically, we first divided the training samples into 10 slices, each corresponding to the local data of one client. We performed a simple experiment on the probability vector $\mathbf{pc} = (\mathrm{pc}_1, \mathrm{pc}_2, \ldots, \mathrm{pc}_K) \sim \mathrm{Dirichlet}(\alpha)$ and assigned the proportion of pc_k of class c instances to clients $k \in \{1, 2, \ldots, K\}$, with $\alpha = 0.01$ to simulate a scenario close to the data heterogeneity in Fig. 3.

Implication 3. *There appears to be a potential constraint effect that can mitigate the dimensional collapse of local models, and this constraint effect is more pronounced among clients with large differences in data distribution. So we need to quantify these differences precisely.*

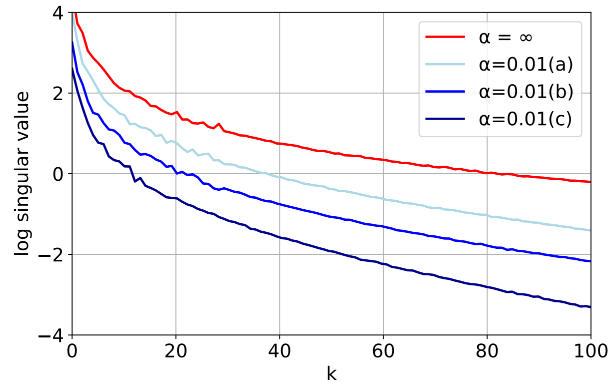

Fig. 3. X-axis(k) is the index of the singular values, and y-axis is the logarithm of the singular values. The red lines represent isomorphic scenarios, and the three lines a,b,c represent paired clients with small and relatively large differences and unpaired, respectively.

D. Promoting Alignment of Model Update Directions Among Participants

To counteract the effects of non-IID data, FedDecorr [25] adds a regularization term to the loss function of each local model to encourage tail singularities not to collapse to 0, thus effectively mitigating dimensional collapse.

Implication 4. *Inspired by FedDecorr, we propose to optimize the training process by integrating a pairwise-based regularization term. The term acts as a mutual restriction on participants with different data distributions, inducing them to try to maintain the same gradient update direction.*

V. Proposed Method

A. Quantify Data Distribution Differences

After recognizing that the key categories of data heterogeneity are label skew and quantity skew, we design a quantitative metric, JSDN. Specifically, this metric consists of two parts: $\mathcal{JSD}$, which indicates the degree of label skew, and N, which indicates the degree of quantity skew.

Jensen-Shannon divergence ($\mathcal{JSD}$) is a measure of similarity between two probability distributions [26]. In the federated learning scenario, we choose this metric to measure the similarity of data labeling distributions between different clients. The $\mathcal{JSD}$ value is negatively correlated with the similarity of data distributions, $\mathcal{JSD} \in [0, 1]$. In general, the data distribution similarity between clients is inversely proportional to the $\mathcal{JSD}$ value. In the extreme case, when the value is 0, it represents that the two sets of data distributions are identical. The value between two distributions (Qa) and (Qb) can be formally defined as:

$$\mathcal{JSD}(Q_a, Q_b) = \frac{1}{2} \sum_{i \in \{a,b\}} \mathcal{KL}(Q_i, \mathbb{M}_{ab}), \tag{9}$$

where Q_a, Q_b are the probability distributions of the data under client a and client b, $\mathcal{KL}(\cdot)$ denotes the KLD (Kullback-Leibler Divergence) [27]. $\mathbb{M}_{ab}$ denotes the mean distribution of Q_a and Q_b:

$$\mathbb{M}_{ab} = \frac{Q_a + Q_b}{2}. \tag{10}$$

N is a quantity skew indicator, representing the difference in data volume distribution among clients. When making predictions on new data, the model may not be able to utilize the knowledge learned from other clients well, leading to a decrease in prediction performance. The data quantity difference between client a and b can be quantified by the following normalized indicator:

$$N = \frac{|N_a - N_b|}{N_a + N_b}, \tag{11}$$

where N_a and N_b are the data volumes of clients a and b, respectively. This equation calculates the normalized value of the difference in data volume between two clients, with $N \in [0, 1]$. The equation for JSDN is given by:

$$JSDN = (1 - \lambda)\mathcal{JSD} + \lambda N. \tag{12}$$

In Eq.(12), λ is a weighting parameter that allows us to adjust the influence of data volume skew relative to label skew. This adjustment is essential, as label skew and volume skew typically have different degrees of impact on the federated learning system's performance. By tuning λ, we can balance the contributions of these two factors in the overall measure of data heterogeneity.

B. Client Pairing Algorithm

1) Data Upload and JSDN Matrix Generation: We assume n clients participate in federated training, each with a dataset corresponding to category C. Before local training begins, each client will first add data perturbations to the local label distribution and data size through LDP, and then upload the processed results to the server. Based on the differential privacy, we have determined a equation that integrates privacy and performance coefficients. The privacy budget allocated to label distribution and data size will be adaptively adjusted according to n and C. The input for the corresponding JSDN calculation becomes the LDP processed $\hat{\mathbf{p}}$ and $\hat{N}$. Specifically, for client i and client j, the calculated JSDN value is denoted as J_{ij}. Combining all the J_{ij} together forms an n-dimensional JSDN matrix. This matrix provides the basis for the subsequent client pairing.

2) Client Pairing Process: When the number of clients n is even, we pair all clients into $n/2$ pairs, starting from the original JSDN matrix that records the JSDN values between all client pairs. We first traverse the JSDN matrix to identify its maximum value J_{max}, with the corresponding row and column subscripts denoted as i and j. This indicates that the data distribution difference between client i and client j is the largest across all pairs. Subsequently, we set all elements in the i-th row, j-th row, i-th column and j-th column of the matrix (excluding the diagonal elements) to zero. For the updated matrix, we again identify the maximum JSDN value and pair the corresponding clients as the second group. This process is repeated iteratively: after each pairing operation, the corresponding rows and columns of the paired clients in the matrix are zeroed out, until all clients are paired. Finally, we obtain $n/2$ client pairs in total. The core idea of this pairing method is to prioritize pairing clients with large data distribution differences together. This is because the constraint relationship between clients with greater distribution differences tends to be stronger.

Algorithm 1: Client Pairing

Input : Label distributions $\hat{\mathbf{p}} = \{\hat{\mathbf{p}}_1, ..., \hat{\mathbf{p}}_n\}$, Data sizes $\hat{N} = \{\hat{N}_1, ..., \hat{N}_n\}$, Clients n
Output: Pairing set P_{pairs}

Initialize $J \leftarrow \text{zeros}(n \times n)$
for $i \leftarrow 1$ **to** n **do**
  for $j \leftarrow i+1$ **to** n **do**
    $J_{ij} = J_{ji} = \text{JSDN}(\hat{\mathbf{p}}_i, \hat{\mathbf{p}}_j; \hat{N}_i, \hat{N}_j)$
  end
end
$P_{\text{pairs}} \leftarrow \emptyset$
if n *even* **then**
  for $k \leftarrow 1$ **to** $n/2$ **do**
    $(i^*, j^*) \leftarrow \arg\max J_{ij}$, $P_{\text{pairs}} \cup = \{(i^*, j^*)\}$
    for $x \leftarrow 1$ **to** n **do**
      $J_{i^*x} = J_{j^*x} = J_{xi^*} = J_{xj^*} = 0$
    end
  end
else
  for $k \leftarrow 1$ **to** $(n-1)/2$ **do**
    $(i^*, j^*) \leftarrow \arg\max J_{ij}$, $P_{\text{pairs}} \cup = \{(i^*, j^*)\}$
    for $x \leftarrow 1$ **to** n **do**
      $J_{i^*x} = J_{j^*x} = J_{xi^*} = J_{xj^*} = 0$
    end
  end
end
return P_{pairs}

When n is odd, we can only form $(n-1)/2$ pairs of clients, and there will be one client left at the end. In this case, the minimum value in the JSDN matrix represents that the data distribution difference between this client and other clients is relatively small. From the perspective of data distribution, this client is roughly in the middle position of the client distribution. That is to say, its label distribution and data volume are representative to a certain extent within the overall client cohort, and its discrepancies from other clients are less pronounced than those between certain other client pairs. Therefore, we choose to abandon the client corresponding to the minimum value in the JSDN matrix to avoid the interference of the parity of the number of federated learning participants on the pairing process, thus simplifying the pairing process and improving the adaptability of the pairing algorithm.

C. FCPC

Motivated by the above observation and analysis of the potential constraints between heterogeneous clients in federated learning, we explore how to fully utilize such constraints to mitigate the impact of data heterogeneity during federation training. During the original federation training, clients train individually with their local datasets and finally upload the

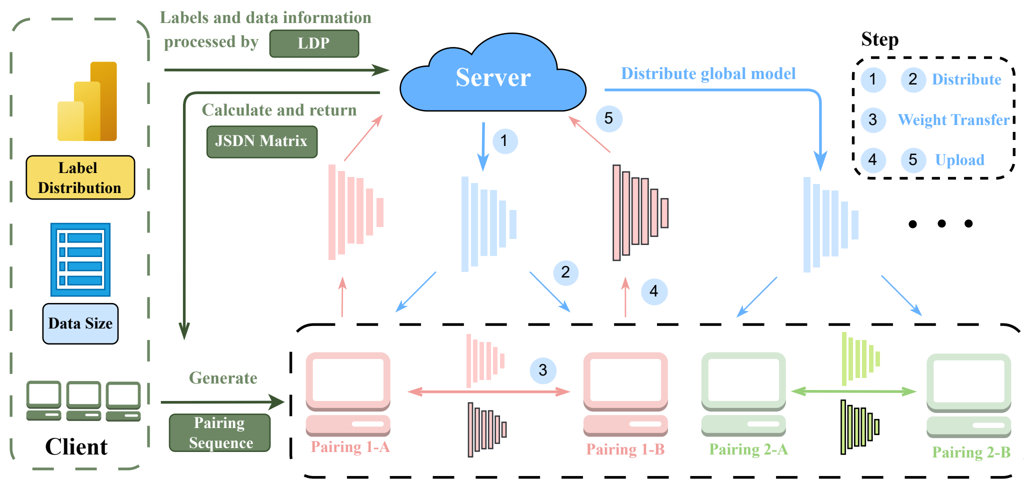

Fig. 4. Framework of FCPC. The left part demonstrates the process that participants in federated learning upload the label distribution and data size to the server after local differential privacy (LDP) processing. The server then calculates the JSDN matrix and returns the pairing sequence. The right part shows the pairing training process of the participants. Taking the red client pair as an example: the server initializes the global model and distributes it to the clients (Steps 1 and 2). At the end of each training round, between the client pairs, the local weights need to be uploaded to the paired client as a constraint term for the client in the next round (Step 3). Meanwhile, the weights are uploaded to the server for global aggregation (Steps 4 and 5).

model weights to the central server for federated averaging, during this period, the clients cannot perceive the data distribution gaps with other clients, and even if mutual constraints exist, they cannot achieve the constraint effect. Therefore, we propose to mitigate this problem by including constraints during local training. A natural way to achieve this is to add the following regularization term to the loss function during training. The system architecture is shown in Fig. 4. Formally, the proposed regularization term is defined as:

$$L_{\text{FCPC}}(w_k, X) = \|w_k^t - w_{k'}^{t-1}\|^2, \tag{13}$$

where w_k^t represents the local weight of client k in t-round training and $w_{k'}^{t-1}$ represents the weight of the client paired with client k in the $(t-1)$-th round of training. At each round, a subset K of the total devices is selected to train. The total communication round is T (As shown in Algorithm 2).

Each client k is selected with probability p_k (a weight coefficient that quantifies the proportional contribution of its local data to the global model), where $p_k \geq 0$ and $\sum_{k=1}^{n} p_k = 1$. This coefficient is calculated as $p_k = N_k / N_{\text{total}}$, with N_k denoting the data volume of client k and $N_{\text{total}} = \sum_{k=1}^{n} N_k$ representing the total data volume across all n clients.

After each round of local training, the clients will send the local training weights to the server and the paired clients. The weights sent to the server are used for global aggregation, and the portion sent to the paired clients is used for constructing constraints for the next round of local training for the paired clients. The overall objective of each local client is

$$\min_{w} \ell(w, X, \mathbf{y}) + \beta L_{\text{FCPC}}(w, X), \tag{14}$$

where ℓ is the cross entropy loss, and β is the regularization coefficient of FCPC. Essentially, L_{FCPC} penalizes the difference between the paired clients, thus preventing the two from going in opposite directions of gradient updating during training. Instead, it constrains the paired clients so that they try to maintain the same direction of gradient updating, speeding up model convergence. This regularization term solves the statistical heterogeneity problem by restricting the local updates of the heterogeneous clients without requiring manually set the number of local epochs. We summarize the steps of FCPC in Algorithm 2.

In addition, detailed convergence analysis of FCPC is provided in the Appendix.

Algorithm 2: FCPC (Proposed Framework)

```
Input  : K, T, β, w^0, n, p_k for k = 1, ..., n
Output: Optimized model w^T
for t ← 0 to T − 1 do
    S_t ← random subset of K devices
    Send w^t to all k ∈ S_t
    foreach device k ∈ S_t do
        Find w_k^{t+1} ≈
          arg min_w [ℓ(w; X, y) + β‖w_k^t − w_{k'}^{t−1}‖^2]
    end
    foreach device k ∈ S_t do
        Send w_k^{t+1} to server
    end
    w^{t+1} ← (1/|S_t|) Σ_{k∈S_t} w_k^{t+1}
end
return w^T
```

VI. Performance Analysis

A. Experimental Setups

In our experimental setup, all participants actively participated in each round of the training process. For network architecture, we used ResNet18 [28] and MobileNetV2 [29], respectively. We performed 100 rounds of communication for all experiments on the CIFAR10/100 [30] datasets. Both CIFAR10 and CIFAR100 have 50, 000 training samples and 10,000 test samples, and the size of each image is 32×32.

Local training was performed for 10 epochs in each round of communication using the SGD optimizer with a learning rate of 0.01 and a batch size of 64. In addition, we applied data enhancement to all experiments.

B. Performance Under Single Types of Non-IID Scenarios

We selected two small and medium-sized datasets, MNIST [31] and FEMNIST, and carried out a detailed study on the accuracy of the FCPC algorithm in single-type non-IID scenarios. During the experiment, we simulated two common single-type non-IID conditions in Table I: label distribution skew and quantity skew.

For the situation of label distribution skew, the results show that FCPC achieves an accuracy of 0.9815 on the MNIST dataset and an accuracy of 0.9734 on the FEMNIST dataset. In comparison, the performance of other methods on these two datasets is relatively weak. This fully demonstrates that when dealing with data with label distribution skew, FCPC can more effectively mine the feature information in the data, improve the model's recognition ability for different types of data, and thus obtain higher accuracy.

TABLE I
TEST ACCURACY ON MNIST AND FEMNIST UNDER SINGLE TYPES OF NON-IID SCENARIOS

Non-IID Scenarios	Dataset	FCPC	FedAvg	FedProx	FedDyn	MOON
Label distribution skew	MNIST	**98.15**	97.32	97.96	97.48	97.54
	FEMNIST	**97.34**	96.35	96.28	97.09	96.56
Quantity skew	MNIST	**98.28**	97.18	97.78	97.63	97.65
	FEMNIST	**97.24**	95.97	96.37	96.52	96.36

In the experimental scenario of quantity skew, FCPC also demonstrates excellent performance. On the MNIST dataset, the accuracy of FCPC reaches 0.9828, and on the FEMNIST dataset, it is 0.9724. It can be seen that when facing the situation of large differences in data quantity, FCPC can still maintain a high accuracy, effectively avoiding the problem of model performance degradation caused by data imbalance.

C. Performance Under Dual Types of Non-IID Scenarios

Addressing mixed types of data skews (e.g., concurrent label and quantity skews) is recognized as a significantly more challenging task in federated learning than handling single-type skews, as the interplay between different heterogeneities can lead to compounded performance degradation. To evaluate the efficacy of FCPC under such complex conditions, experiments were conducted on the CIFAR-10 dataset under dual-skew scenarios. The degree of heterogeneity was controlled using a Dirichlet distribution parameter $\alpha \in \{0.05, 0.1, 0.5\}$, simulating environments with extreme, high, and moderate data heterogeneity, respectively.

Experimental results, summarized in Table II, demonstrate that the proposed FCPC framework consistently achieves superior classification accuracy across all tested heterogeneity levels. In contrast, the performance of all baseline methods exhibits a more substantial decline, particularly under the extreme heterogeneity condition ($\alpha = 0.05$).

TABLE II
PERFORMANCE COMPARISON UNDER DIFFERENT LEVELS OF DUAL-SKEWED NON-IID DATA ON CIFAR-10 AND CIFAR-100

Dataset	CIFAR10			CIFAR100		
Method / α	0.05	0.1	0.5	0.05	0.1	0.5
FedAvg (AISTATS, 2017)	53.74	65.19	79.98	49.76	56.43	61.40
FedProx (MLsys, 2020)	53.21	65.20	79.46	50.22	56.37	61.59
MOON (CVPR, 2021)	59.68	67.60	80.04	45.68	56.37	61.81
FedDyn (ICPADS, 2023)	62.24	68.49	77.38	49.73	56.36	61.28
FBLG (IJCAI, 2024)	62.95	69.13	79.82	50.45	57.62	61.38
FedCFA(AAAI, 2025)	63.15	69.83	80.45	51.47	57.81	62.13
FCPC	**63.17**	**70.59**	**80.28**	**52.53**	**58.12**	**62.72**

D. Orthogonality of FCPC with Existing FL Algorithms

Thanks to the regularization property of FCPC method, it acts on the local training stage through pairing between devices. It only needs to introduce the pairing algorithm on the basis of other existing methods and add an additional term L_{FCPC} on the loss function. In addition, it does not need any change, which makes FCPC have the natural advantage of plug-and-play.

To validate this orthogonality, FCPC is integrated into five representative baseline algorithms, namely FedAvg [22], FedProx [14], MOON [15], FBLG [19], and FedCFA [20]. Two sets of experiments are conducted: (1) CIFAR-10 dataset with the ResNet-18 model, where data are partitioned among 10 clients; (2) TinyImageNet dataset with the same client partitioning strategy. The degree of heterogeneity is quantified using a Dirichlet distribution with $\alpha \in \{0.05, 0.1, 0.5\}$ (smaller α indicates stronger heterogeneity). The experimental objective is to verify whether integrating FCPC can enhance the performance of baseline methods under varying non-IID degrees.

Experimental results indicate that embedding FCPC as a regularization term into baseline algorithms yields notable performance gains, especially under strongly heterogeneous settings ($\alpha = 0.05$ and $\alpha = 0.1$). On the CIFAR-10 dataset, the performance improvement ranges from approximately 4.0% to 9.4% compared with the original baselines. For instance, FedAvg achieves 53.74% accuracy when $\alpha = 0.05$, and the accuracy increases to 63.17% after integrating FCPC, representing a 9.43% improvement. On the TinyImageNet dataset, the improvement under $\alpha = 0.05$ and $\alpha = 0.1$ is about 5.0% and 3.5%–4.5%, respectively.

The results presented in Tables III and IV collectively demonstrate the effectiveness and orthogonality of integrating FCPC with existing FL algorithms. Notably, FCPC consistently enhances the performance of baseline methods across different datasets and heterogeneity levels, which verifies its ability to strengthen the non-IID data handling capability of FL frameworks.

TABLE III
ORTHOGONALITY ANALYSIS OF FCPC INTEGRATION WITH EXISTING FL ALGORITHMS (RESNET-18, CIFAR10)

α	FedAvg (AISTATS 2017)		FedProx (MLsys 2020)		MOON (CVPR 2021)		FBLG (IJCAI 2024)		FedCFA (AAAI 2025)	
	Orig.	**+FCPC**	Orig.	**+FCPC**	Orig.	**+FCPC**	Orig.	**+FCPC**	Orig.	**+FCPC**
0.05	53.74	63.17	53.21	61.29	59.68	63.46	62.95	63.08	63.15	**64.23**
0.1	65.19	70.59	65.20	**72.83**	67.60	72.26	69.13	70.45	69.83	72.29
0.5	79.98	80.28	79.46	80.35	80.04	80.82	79.82	80.83	80.45	**81.62**

TABLE IV
ORTHOGONALITY ANALYSIS OF FCPC INTEGRATION WITH EXISTING FL ALGORITHMS (TINYIMAGENET)

Method	TinyImageNet		
	$\alpha = 0.05$	$\alpha = 0.1$	$\alpha = 0.5$
FedAvg (AISTATS 2017)	35.48	39.53	46.97
+FCPC	40.47	44.36	50.28
FedProx (MLsys 2020)	35.50	39.69	47.23
+FCPC	40.68	44.33	50.53
MOON (CVPR 2021)	35.49	40.81	47.91
+FCPC	40.64	44.42	51.32
FedCFA (AAAI 2025)	35.62	40.94	48.11
+FCPC	40.71	44.57	51.48

E. Ablation Study

To deeply explore the impact of the key hyperparameters in the FCPC algorithm and the selection of the pairing indicators of the federated learning participants on the model performance, we carried out comprehensive ablation experiments, focusing on the parameter λ, the parameter β, and the similarity measure. These key parameters play a crucial role in the FCPC algorithm, directly affecting the model's processing effect on data heterogeneity and the final performance.

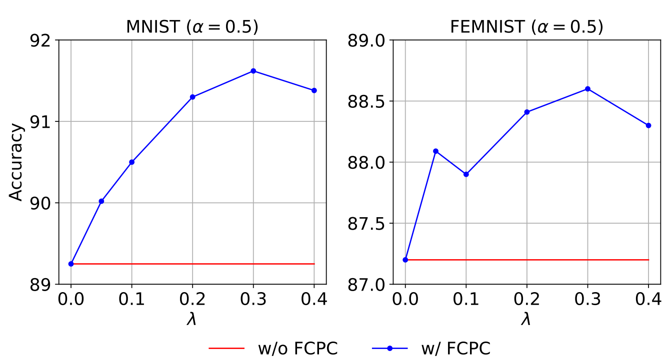

Fig. 5. Ablation studies of $\boldsymbol{\lambda}$. We set the value of $\boldsymbol{\alpha}$ to 0.5 and select the MNIST and FEMNIST datasets to conduct experiments with $\boldsymbol{\lambda}$ taking values from the set $\{\mathbf{0.1}, \mathbf{0.2}, \mathbf{0.3}, \mathbf{0.4}\}$ to explore the appropriate value of $\boldsymbol{\lambda}$.

Coefficient λ**.** Considering the characteristic of the FedAvg algorithm itself, which performs weighted averaging based on the amount of data, the label skew may play a dominant role. Based on this, we search for the relatively most appropriate value of λ in the set $\{0.1, 0.2, 0.3, 0.4\}$ to measure the distribution difference between clients. As λ gradually increases, the model's adaptability to the data amount difference is enhanced. However, when λ is too large, it may overemphasize the data amount deviation and relatively weaken the impact of the label deviation, causing the model performance to be affected again. After a series of experimental comparisons, it is found that when $\lambda = 0.3$, the JSDN indicator can more effectively measure the distribution difference of client data, enabling the FCPC algorithm to achieve good performance in various experimental scenarios. Meanwhile, we also conducted a rigorous theoretical derivation of λ values in the Appendix. As shown in Fig. 5, it is recommended to set λ to 0.3 to fully utilize the advantages of the FCPC algorithm in handling dual-skewed non-IID data.

Coefficient β**.** We systematically evaluated the impact of the hyperparameter β on FCPC's performance across various data-heterogeneity scenarios (controlled by $\alpha \in \{0.05, 0.1\}$). Experiments were conducted with $\beta \in \{0.05, 0.1, 0.2, 0.3\}$ while keeping all other conditions (network architecture, dataset, training epochs, optimizer settings, and data augmentation) identical. The results show that FCPC's performance improves with increasing β, stabilizes, and then slightly declines. Moderate β values strengthen regularization, enhancing inter-client gradient consistency and improving performance. Conversely, excessive β introduces overly strong constraints that hinder learning, leading to suboptimal performance. Across all experimental configurations, $\beta = 0.2$ consistently yielded near-optimal results (see Fig. 6), representing an ideal balance between constraint strength and learning capacity. This value serves as a robust default when prior information is unavailable.

TABLE V
PERFORMANCE COMPARISON OF DIFFERENT DISSIMILARITY METRICS AT $\boldsymbol{\alpha = 0.5}$. JSDN REPRESENTS THE COMPREHENSIVE INDICATOR OF JENSEN-SHANNON DIVERGENCE AND DATA SIZE.

Dataset	Dissimilarity Metrics			
	JSDN	Jensen-Shannon divergence	Euclidean distance	Cosine distance
MNIST	**91.63**	90.52	90.27	90.38
FEMNIST	**88.61**	87.47	86.30	86.45
CIFAR10	**80.08**	78.52	76.75	76.70

Dissimilarity measures. We first validate the effect of the JSDN metric adopted in this paper on the performance of FCPC at $\alpha = 0.5$, and then replace the dissimilarity metrics with Jensen-Shannon divergence, cosine distance, and Euclidean distance, respectively, and then present the classification accuracy of the four metrics in a table, as shown

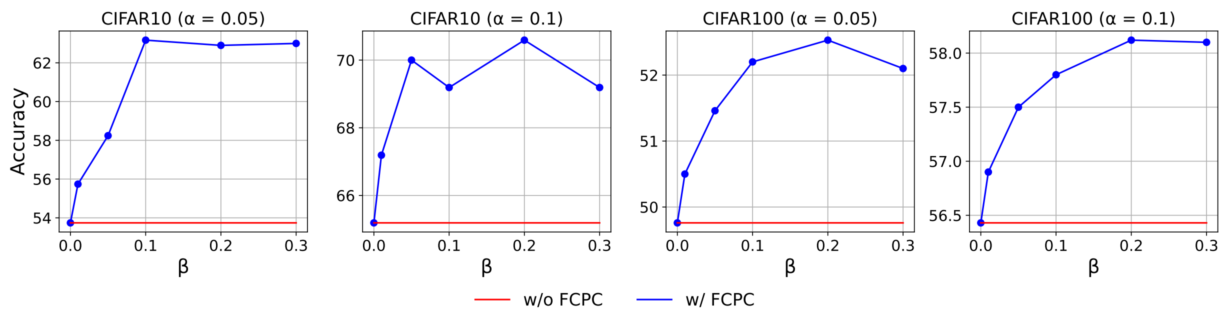

Fig. 6. **Ablation studies on β.** We select a value of β that is as adaptive as possible to multiple scenarios by comparing the effect of different regularization strengths on model performance on FedAvg.

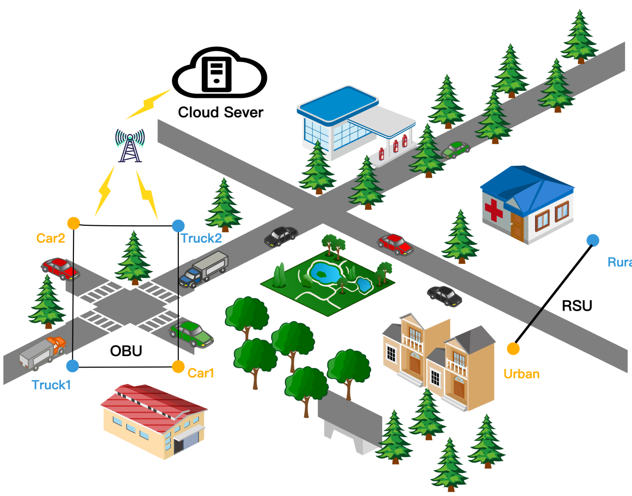

Fig. 7. The heterogeneous information sources in urban and rural areas form the RSU, while different modes of transportation such as trucks and cars form the OBU, collectively forming the heterogeneous data network of the IoT.

in Table V. We observe that our proposed FCPC performs best on the JSDN dissimilarity metric. This preference is attributed to the unique advantage of JSDN to measure both label distribution and quantity distribution.

F. Engineering Applications

In intelligent transportation systems, the collaborative optimization of traffic signal timing and vehicle route planning relies heavily on the fusion of multi-source IoT data, among which the vehicle-road collaborative network composed of roadside units (RSUs), on-board units (OBUs), and traffic light controllers is a typical application scenario. These IoT devices are distributed in diverse road segments and generate a large amount of non-IID traffic data due to differences in traffic flow density, vehicle types, and travel purposes: for example, RSUs at urban intersections in peak hours mainly collect data with high congestion labels and large sample sizes, while RSUs in suburban roads mostly obtain data with low congestion labels and small sample sizes; OBUs of freight vehicles on highways generate data with stable high-speed characteristics, which are significantly different from the data with frequent start-stop characteristics generated by OBUs of passenger cars in urban areas.

To address this, the FCPC algorithm can be integrated into the intelligent transportation IoT system (see Fig. 8): each RSU, OBU, and traffic light controller acts as a client in federated learning, and first uploads the label distribution and data size of local traffic data to the cloud server after LDP processing; the server calculates the JSDN matrix based on the uploaded information, and pairs clients with significant differences in data distribution through the client pairing algorithm; during local training, the paired clients constrain each other through the FCPC regularization term, which reduces the divergence of gradient update directions caused by non-IID data, enables the local models to learn the characteristics of heterogeneous traffic data more comprehensively, and then aggregates the local model parameters to form a global model. This global model can be used to predict short-term traffic flow in each road segment and optimize traffic signal timing dynamically, thereby reducing traffic congestion and improving travel efficiency.

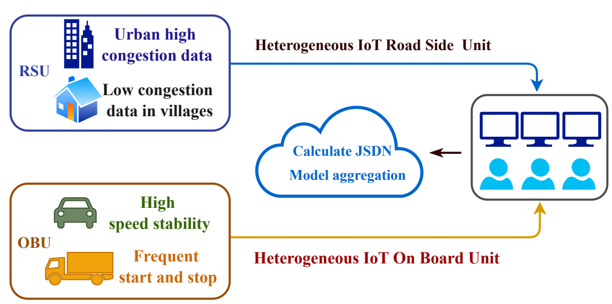

Fig. 8. **FCPC in IoT.** This diagram illustrates the engineering application logic of the FCPC framework in IoT-driven intelligent transportation systems.

In practical applications, the plug-and-play nature of FCPC allows new IoT devices to be seamlessly integrated into the federated learning system without modifying the underlying framework, which is highly adaptable to the continuous expansion of intelligent transportation infrastructure. Our subsequent work will focus on applying the FCPC algorithm to the collaborative response of unexpected traffic events, exploring how to use the mutual constraint mechanism between heterogeneous clients to improve the real-time performance and accuracy of event handling, and further enhancing the intelligence level of the transportation system.

VII. Conclusion

This paper addresses the critical challenge of performance degradation in federated learning (FL) caused by dual-skewed (label and quantity) non-IID data in Internet of Things (IoT) systems. A key insight driving this work is that the mutual constraint potential between clients is positively correlated with their data distribution dissimilarity. Building on this insight, a novel, dissimilarity-aware framework named FCPC is proposed, which comprises two synergistic components: (1) the Jensen-Shannon Divergence and Skew Indicator Number (JSDN) index for precise client heterogeneity quantification, and (2) a client pairing and mutual-constraint mechanism that employs a lightweight regularization term during local training to align gradient updates and mitigate dimensional collapse.

Extensive experiments on multiple datasets (CIFAR-10/100, TinyImageNet, MNIST, FEMNIST) and network architectures (ResNet-18, MobileNetV2) confirm FCPC's efficacy. It consistently outperforms strong baselines (FedAvg, FedProx, MOON), achieving top-1 accuracy on CIFAR-10 under severe dual-skew conditions. Crucially, FCPC's plug-and-play design enables seamless integration into existing FL frameworks without introducing prohibitive computational burden, offering a practical system integration strategy for resource-constrained IoT environments.

Future research will explore: (1) adaptive mechanisms for dynamic parameter tuning in evolving IoT environments; (2) extension to broader heterogeneity types (e.g., feature skew and concept drift); and (3) validation in cross-silo industrial FL scenarios with stringent privacy requirements.

References

[1] B. Liu, N. Lv, Y. Guo, and Y. Li, "Recent advances on federated learning: A systematic survey," *Neurocomputing*, p. 128019, 2024.

[2] Z. An, X. Yu, C. Wang, Y. Zhang, and C. Song, "Embodied intelligence: Recent advances and future perspectives," *The Innovation Informatics*, vol. 1, no. 1, pp. 100 008–1, 2025.

[3] X. Wang, X. Liang, and N. Wu, "A neural network-assisted boussinesq pressure bulb model for load assessment of cylindrical rolling element bearings," *Mech. Syst. Signal Process.*, vol. 243, p. 113655, 2026.

[4] J. Zhuang, J. Wu, G. Pei, W. Li, G. Xin, C. Ma, K. Feng, L. Zhang, and C. Yang, "A neuro-fuzzy approach with hypergraph convolution for fault diagnosis in industrial devices," *J. Reliab. Sci. Eng.*, vol. 1, no. 3, p. 035301, 2025.

[5] K. Feng, J. Ji, Q. Ni, and M. Beer, "A review of vibration-based gear wear monitoring and prediction techniques," *Mech. Syst. Signal Process.*, vol. 182, p. 109605, 2023.

[6] C. Dai, S. Lu, P. Wang, H. Yin, Y. Wang, X. Liu, and B. Guo, "Client selection in federated learning for industry 5.0: A heuristic-guided pointer network reinforcement learning approach," *IEEE Trans. Ind. Informatics*, pp. 1–11, 2025.

[7] Z. Tong, J. Wang, X. Hou, J. Chen, Z. Jiao, and J. Liu, "Blockchain-based trustworthy and efficient hierarchical federated learning for uav-enabled iot networks," *IEEE Internet Things J.*, vol. 11, no. 21, pp. 34 270–34 282, 2024.

[8] S. Xu, J. Zou, C. Zhou, H. Wu, and Z. Zeng, "Spatio–temporal aware personalized federated learning for load forecasting in power systems," *IEEE Trans. Ind. Informatics*, vol. 21, no. 8, pp. 6505–6515, 2025.

[9] F. Almalik, N. Alkhunaizi, I. Almakky, and K. Nandakumar, "Fesvibs: Federated split learning of vision transformer with block sampling," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assisted Intervention (MICCAI)*. Springer, 2023, pp. 350–360.

[10] C. Thapa, P. C. M. Arachchige, S. Camtepe, and L. Sun, "Splitfed: When federated learning meets split learning," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 8, 2022, pp. 8485–8493.

[11] D. Deng, W. Zhao, X. Wu, T. Zhang, J. Zheng, J. Kang, and D. Niyato, "Decffd: A personalized federated learning framework for cross-location fault diagnosis," *IEEE Trans. Ind. Informatics*, vol. 20, no. 5, pp. 7082–7091, 2024.

[12] J. Tang, K. Fan, W. Xie, F. Han, Z. Qu, A. Liu, N. N. Xiong, S. Zhang, and T. Wang, "Btv-cmab: A bi-directional trust verification-based combinatorial multiarmed bandit scheme for mobile crowdsourcing," *IEEE Internet Things J.*, vol. 11, no. 2, pp. 1925–1938, 2023.

[13] S. Guo, H. Chen, Y. Liu, C. Yang, Z. Li, and C. H. Jin, "Heterogeneous federated learning framework for iiot based on selective knowledge distillation," *IEEE Trans. Ind. Informatics*, vol. 21, no. 2, pp. 1078–1089, 2025.

[14] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," *Proc. Mach. Learn. Syst.*, vol. 2, pp. 429–450, 2020.

[15] Q. Li, B. He, and D. Song, "Model-contrastive federated learning," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 10 713–10 722.

[16] K. Cao, Y. Cui, R. Xu, Y. Sun, Z. Liu, and C. Tan, "Personalized federated learning for green industrial iot," *IEEE Trans. Ind. Informatics*, 2025.

[17] C. Jin, X. Chen, Y. Gu, and Q. Li, "Feddyn: A dynamic and efficient federated distillation approach on recommender system," in *2022 IEEE 28th international conference on parallel and distributed systems (ICPADS)*. IEEE, 2023, pp. 786–793.

[18] J. Guo, Z. Liu, S. Tian, F. Huang, J. Li, X. Li, K. K. Igorevich, and J. Ma, "Tfl-dt: A trust evaluation scheme for federated learning in digital twin for mobile networks," *IEEE Journal on Selected Areas in Communications*, vol. 41, no. 11, pp. 3548–3560, 2023.

[19] Y. Xu, Y. Li, H. Luo, X. Fan, and X. Liu, "Fblg: A local graph based approach for handling dual skewed non-iid data in federated learning," in *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, IJCAI-24, K. Larson, Ed. International Joint Conferences on Artificial Intelligence Organization*, vol. 8, 2024, pp. 5289–5297.

[20] Z. Jiang, J. Xu, S. Zhang, T. Shen, J. Li, K. Kuang, H. Cai, and F. Wu, "Fedcfa: Alleviating simpson's paradox in model aggregation with counterfactual federated learning," in *Proc. AAAI Conf. Artif. Intell.*, vol. 39, no. 17, 2025, pp. 17 662–17 670.

[21] S. Wang, Y. Tian, J. Xiong, J. Ma, and Y. Zhang, "Verifydfl: Secure aggregation for decentralized federated learning with input validation in mobile edge intelligence," *IEEE Transactions on Cognitive Communications and Networking*, 2025.

[22] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Artificial intelligence and statistics*. PMLR, 2017, pp. 1273–1282.

[23] C. Dwork, F. McSherry, K. Nissim, and A. Smith, "Calibrating noise to sensitivity in private data analysis," in *Theory of Cryptography: Third Theory of Cryptography Conference, TCC 2006, New York, NY, USA, March 4-7, 2006. Proceedings 3*. Springer, 2006, pp. 265–284.

[24] T. Wang, J. Blocki, N. Li, and S. Jha, "Locally differentially private protocols for frequency estimation," in *26th USENIX Security Symposium (USENIX Security 17)*, 2017, pp. 729–745.

[25] Y. Shi, J. Liang, W. Zhang, V. Y. Tan, and S. Bai, "Towards understanding and mitigating dimensional collapse in heterogeneous federated learning," in *ICLR*, 2023.

[26] J. Lin, "Divergence measures based on the shannon entropy," *IEEE Trans. Inf. Theory*, vol. 37, no. 1, pp. 145–151, 1991.

[27] T. Van Erven and P. Harremos, "Rényi divergence and kullback-leibler divergence," *IEEE Transactions on Information Theory*, vol. 60, no. 7, pp. 3797–3820, 2014.

[28] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.

[29] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "Mobilenetv2: Inverted residuals and linear bottlenecks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 4510–4520.

[30] A. Krizhevsky, G. Hinton *et al.*, "Learning multiple layers of features from tiny images," 2009.

[31] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.